

22. I/O Ports

22.1 Overview

The pins of an I/O port function as general I/O port pins, I/O pins for peripheral modules, interrupt input pins, or bus control pins.

Each pin is also configurable as an I/O pin of a peripheral module or an input pin for an interrupt. All pins function as input pins immediately after a reset, and pin functions are switched by register settings. The setting of each pin is specified by the registers for the corresponding I/O port and peripheral modules.

Each port has the port direction register (PDR) that selects input or output direction, the port output data register (PODR) that holds data for output, the port input register (PIDR) that indicates the pin states, the open-drain control register y (ODRy, y = 0, 1) that selects the output type of each pin, the pull-up resistor control register (PCR) that controls on/off of the input pull-up resistors, the drive capacity control register (DSCR) that selects the drive capacity, and the port mode register (PMR) that specifies the pin function of each port.

For details on PMR, refer to section 23, Multi-Function Pin Controller (MPC).

The configuration of the I/O ports differs depending on the package. Table 22.1 lists the specifications of I/O ports, and Table 22.2 lists the port functions.

Table 22.1 Specifications of I/O Ports

Port Symbol	Package		Package		Package		Package	
	224 Pins	Number of Pin	176 Pins	Number of Pin	145 or 144 Pins	Number of Pin	100 Pins	Number of Pin
PORT0	P00 to P03, P05, P07	6	P00 to P03, P05, P07	6	P00 to P03, P05, P07	6	P05, P07	2
PORT1	P10 to P17	8	P10 to P17	8	P12 to P17	6	P12 to P17	6
PORT2	P20 to P27	8	P20 to P27	8	P20 to P27	8	P20 to P27	8
PORT3	P30 to P37	8	P30 to P37	8	P30 to P37	8	P30 to P37	8
PORT4	P40 to P47	8	P40 to P47	8	P40 to P47	8	P40 to P47	8
PORT5	P50 to P57	8	P50 to P57	8	P50 to P56	7	P50 to P55	6
PORT6	P60 to P67	8	P60 to P67	8	P60 to P67	8	Not provided	0
PORT7	P70 to P77	8	P70 to P77	8	P70 to P77	8	Not provided	0
PORT8	P80 to P87	8	P80 to P87	8	P80 to P83, P86, P87	6	Not provided	0
PORT9	P90 to P97	8	P90 to P97	8	P90 to P93	4	Not provided	0
PORTA	PA0 to PA7	8	PA0 to PA7	8	PA0 to PA7	8	PA0 to PA7	8
PORTB	PB0 to PB7	8	PB0 to PB7	8	PB0 to PB7	8	PB0 to PB7	8
PORTC	PC0 to PC7	8	PC0 to PC7	8	PC0 to PC7	8	PC0 to PC7	8
PORTD	PD0 to PD7	8	PD0 to PD7	8	PD0 to PD7	8	PD0 to PD7	8
PORTE	PE0 to PE7	8	PE0 to PE7	8	PE0 to PE7	8	PE0 to PE7	8
PORTF	PF0 to PF5	6	PF0 to PF5	6	PF5	1	Not provided	0
PORTG	PG0 to PG7	8	PG0 to PG7	8	Not provided	0	Not provided	0
PORTH	PH0 to PH7	8	Not provided	0	Not provided	0	Not provided	0
PORTJ	PJ0 to PJ3, PJ5	5	PJ0 to PJ3, PJ5	5	PJ3, PJ5	2	PJ3	1
PORTK	PK0 to PK7	8	Not provided	0	Not provided	0	Not provided	0
PORTL	PL0 to PL7	8	Not provided	0	Not provided	0	Not provided	0
PORTM	PM0 to PM7	8	Not provided	0	Not provided	0	Not provided	0
PORTN	PN0 to PN5	6	Not provided	0	Not provided	0	Not provided	0
PORTQ	PQ0 to PQ7	8	Not provided	0	Not provided	0	Not provided	0
Total of pins		183	Total of pins		137	Total of pins		112
								79

Table 22.2 Port Functions (1/2)

Port Symbol	Pin	Input Pull-up	Open-Drain Output	Driving Ability Switching	5-V Tolerant
PORT0	P00 to P02	✓	✓	Normal drive/high drive/high-speed interface high-drive	—
	P03, P05	✓	✓	Fixed to high driving ability output	—
	P07	✓	✓	Fixed to high driving ability output	✓
PORT1	P10	✓	✓	Fixed to high driving ability output	—
	P11 to P14	✓	✓	Normal drive/high drive/high-speed interface high-drive	✓
	P15, P16	✓	✓	Fixed to high driving ability output	✓
	P17	✓	✓	High drive/high-speed interface high-drive	✓
PORT2	P20, P21	✓	✓	High drive/high-speed interface high-drive	✓
	P22, P23	✓	✓	High drive/high-speed interface high-drive	—
	P24 to P26	✓	✓	Fixed to high driving ability output	—
	P27	✓	✓	Normal drive/high drive/high-speed interface high-drive	—
PORT3	P30, P31	✓	✓	High drive/high-speed interface high-drive	✓
	P32, P33	✓	✓	Fixed to high driving ability output	✓
	P34, P37	✓	✓	Fixed to high driving ability output	—
	P35	—	—	—	—
	P36	✓	✓	Fixed to Normal driving ability output	—
PORT4	P40 to P47	✓	✓	Fixed to Normal driving ability output	—
PORT5	P50 to P52, P54 to P57	✓	✓	Normal drive/high drive/high-speed interface high-drive	—
	P53	✓	✓	High drive/high-speed interface high-drive	—
PORT6	P60 to P66	✓	✓	High drive/high-speed interface high-drive	—
	P67	✓	✓	High drive/high-speed interface high-drive	✓
PORT7	P70, P73	✓	✓	High drive/high-speed interface high-drive	—
	P71	✓	✓	Fixed to high driving ability output	—
	P72, P74 to P77	✓	✓	Normal drive/high drive/high-speed interface high-drive	—
PORT8	P80 to P85	✓	✓	Normal drive/high drive/high-speed interface high-drive	—
	P86	✓	✓	Fixed to high driving ability output	—
	P87	✓	✓	High drive/high-speed interface high-drive	—
PORT9	P90 to P97	✓	✓	Normal drive/high drive/high-speed interface high-drive	—
PORTA	PA0 to PA7	✓	✓	Normal drive/high drive/high-speed interface high-drive	—
PORTB	PB0 to PB7	✓	✓	Normal drive/high drive/high-speed interface high-drive	—
PORTC	PC0 to PC3	✓	✓	Normal drive/high drive/high-speed interface high-drive	✓
	PC4 to PC7	✓	✓	Normal drive/high drive/high-speed interface high-drive	—
PORTD	PD0 to PD7	✓	✓	Normal drive/high drive/high-speed interface high-drive	—
PORTE	PE0 to PE7	✓	✓	Normal drive/high drive/high-speed interface high-drive	—
PORTF	PF0 to PF5	✓	✓	Fixed to high driving ability output	—

Table 22.2 Port Functions (2/2)

Port Symbol	Pin	Input Pull-up	Open-Drain Output	Driving Ability Switching	5-V Tolerant
PORTG	PG0, PG1	✓	✓	Normal drive/high drive/high-speed interface high-drive	—
	PG2 to PG7	✓	✓	High drive/high-speed interface high-drive	—
PORTH	PH0 to PH7	✓	✓	Normal drive/high drive/high-speed interface high-drive	—
PORTJ	PJ0 to PJ2	✓	✓	Normal drive/high drive/high-speed interface high-drive	—
	PJ3, PJ5	✓	✓	Fixed to high driving ability output	—
PORTK	PK0 to PK7	✓	✓	Normal drive/high drive/high-speed interface high-drive	—
PORTL	PL0 to PL7	✓	✓	Normal drive/high drive/high-speed interface high-drive	—
PORTM	PM0 to PM7	✓	✓	Normal drive/high drive/high-speed interface high-drive	—
PORTN	PN0 to PN5	✓	✓	Normal drive/high drive/high-speed interface high-drive	—
PORTQ	PQ0 to PQ7	✓	✓	Normal drive/high drive/high-speed interface high-drive	—

Specifying input pull-up, open-drain output, switching of driving ability, or 5-V tolerance is available for other signals on pins that also function as general I/O pins.

22.2 I/O Port Configuration

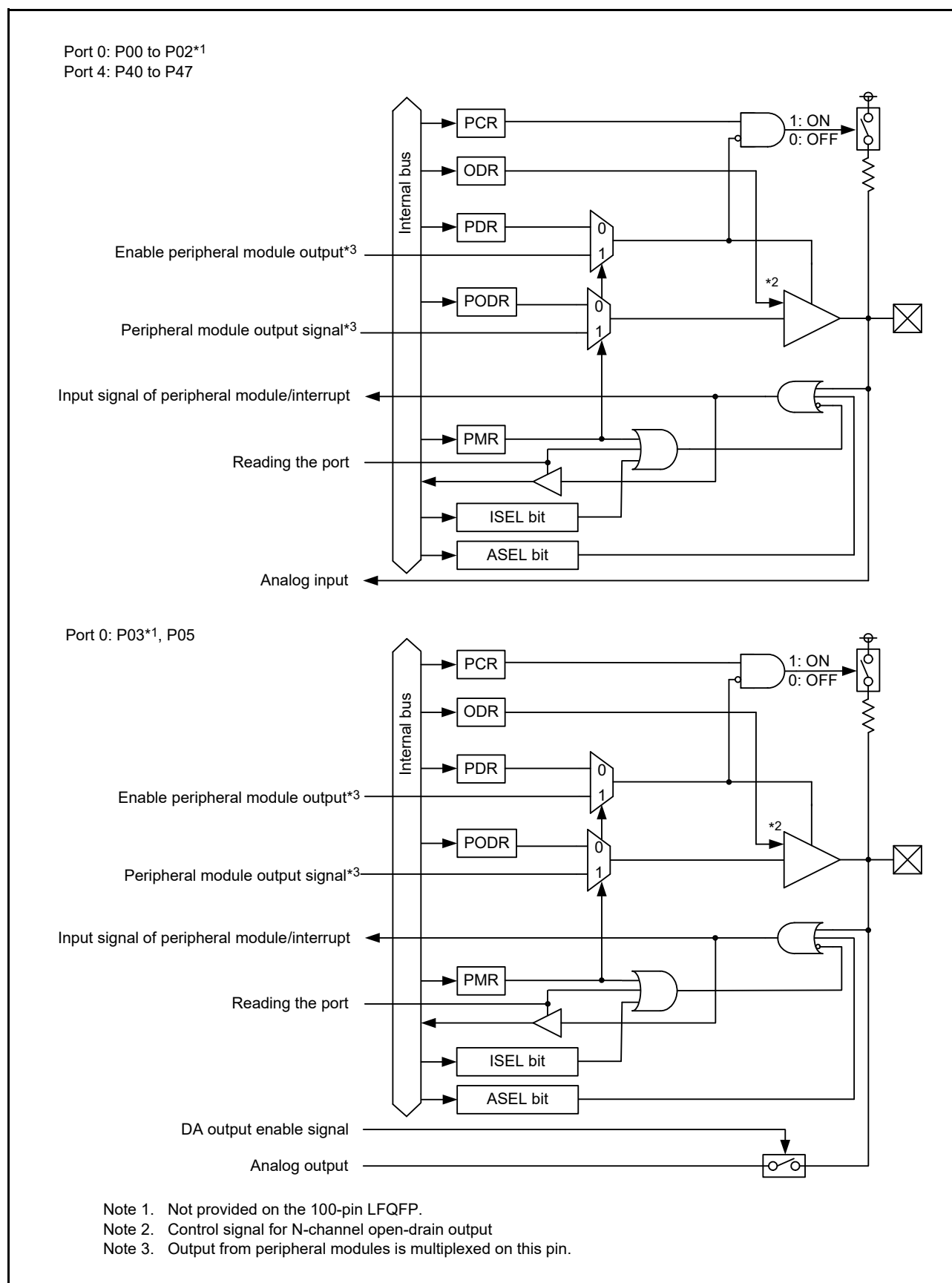
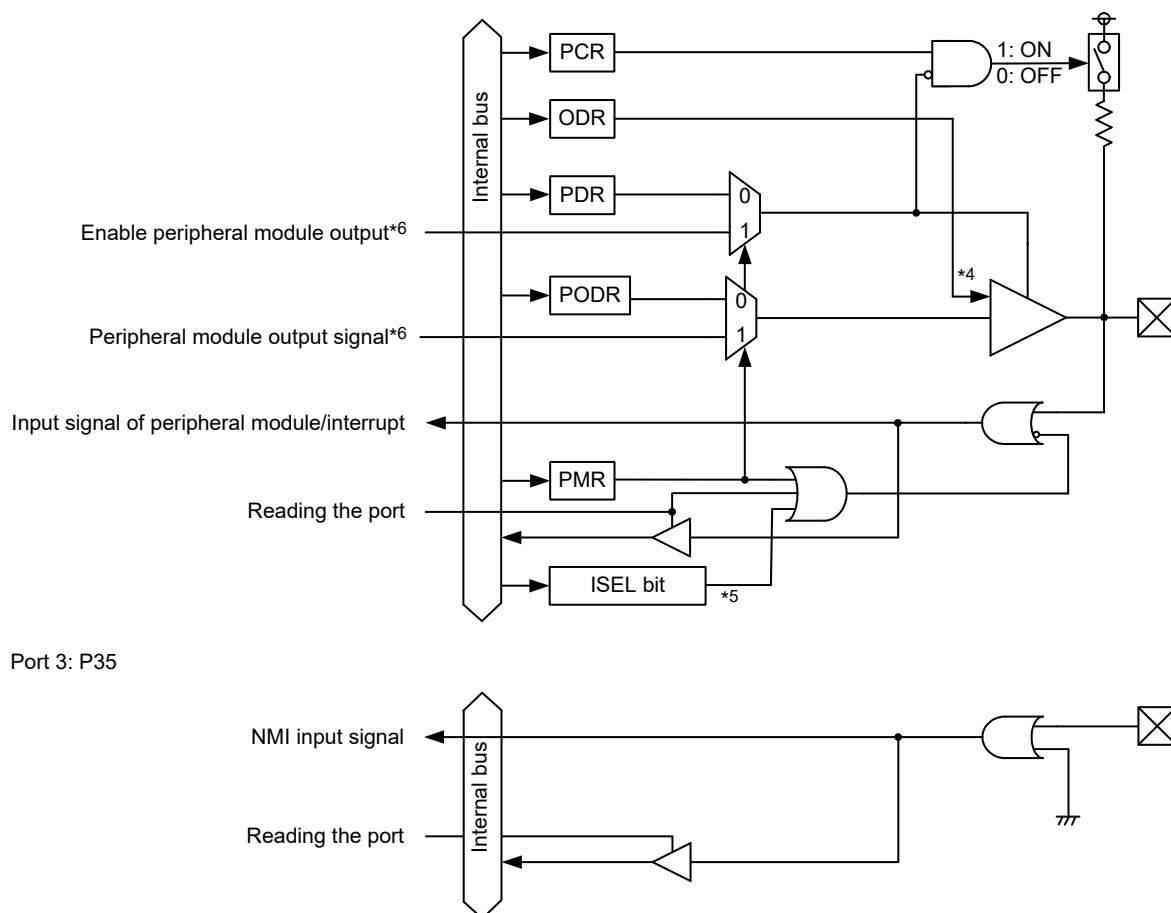


Figure 22.1 I/O Port Configuration (1)

Port 0: P07
 Port 1: P11*2, *3, P14 to P17
 Port 2: P20 to P23
 Port 3: P30 to P34, P36, P37
 Port 5: P56*3, P57*2, *3
 Port 8: P80 to P83*3, P84*2, *3, P85*2, *3, P86*3, P87*3
 Port F: PF0 to PF4*2, *3, PF5*3
 Port H: PH0 to PH7*1, *2, *3
 Port J: PJ0 to PJ2*2, *3, PJ3, PJ5*3
 Port K: PK0 to PK7*1, *2, *3
 Port L: PL0 to PL7*1, *2, *3
 Port M: PM0 to PM7*1, *2, *3
 Port N: PN0 to PN5*1, *2, *3
 Port Q: PQ0 to PQ7*1, *2, *3



- Note 1. Not provided on the 176-pin LFBGA and 176-pin LFQFP.
 Note 2. Not provided on the 145-pin TFLGA and 144-pin LFQFP.
 Note 3. Not provided on the 100-pin LFQFP.
 Note 4. Control signal for N-channel open-drain output
 Note 5. An external interrupt function is multiplexed on this pin.
 Note 6. Output from peripheral modules is multiplexed on this pin.

Figure 22.2 I/O Port Configuration (2)

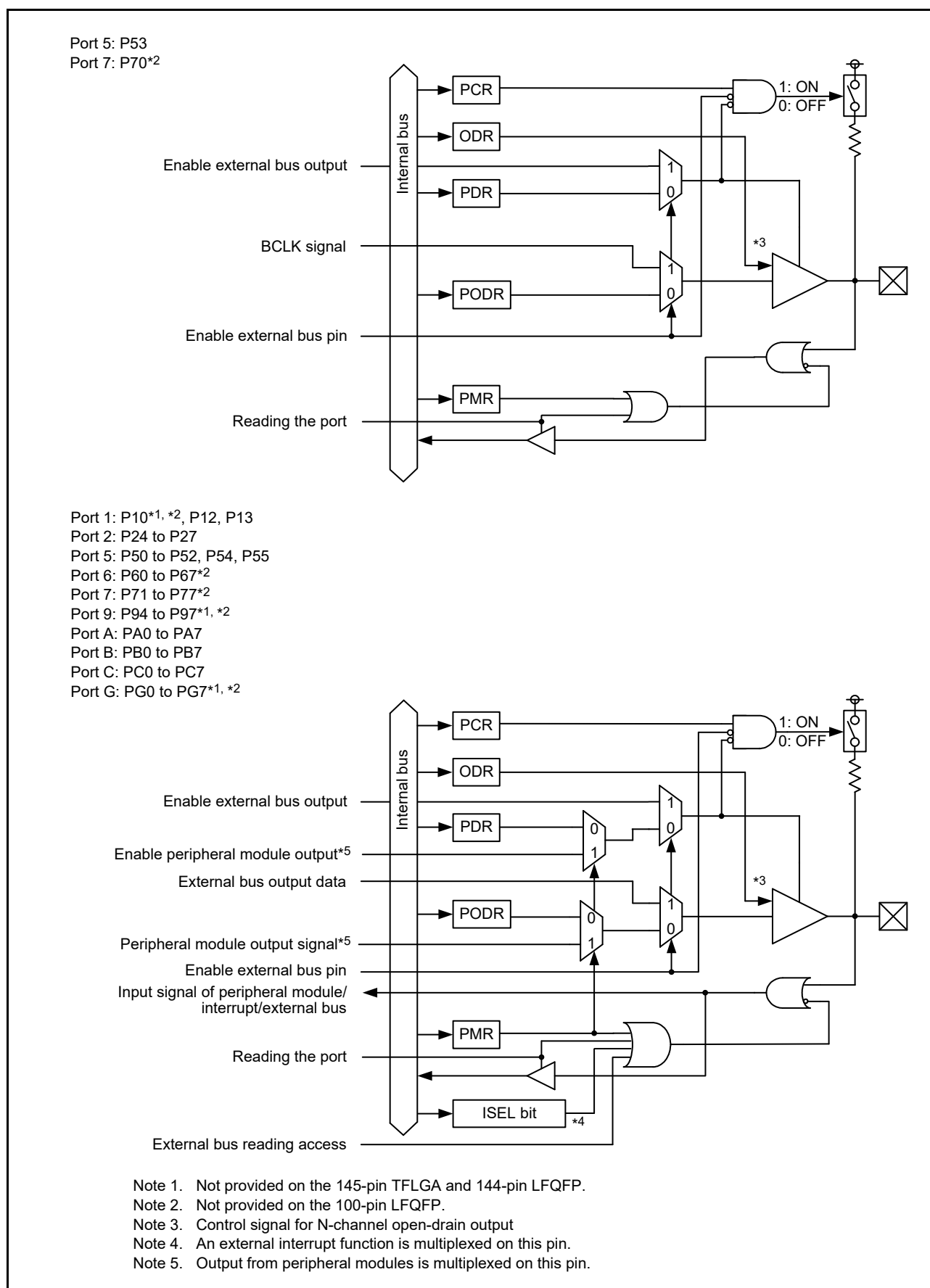


Figure 22.3 I/O Port Configuration (3)

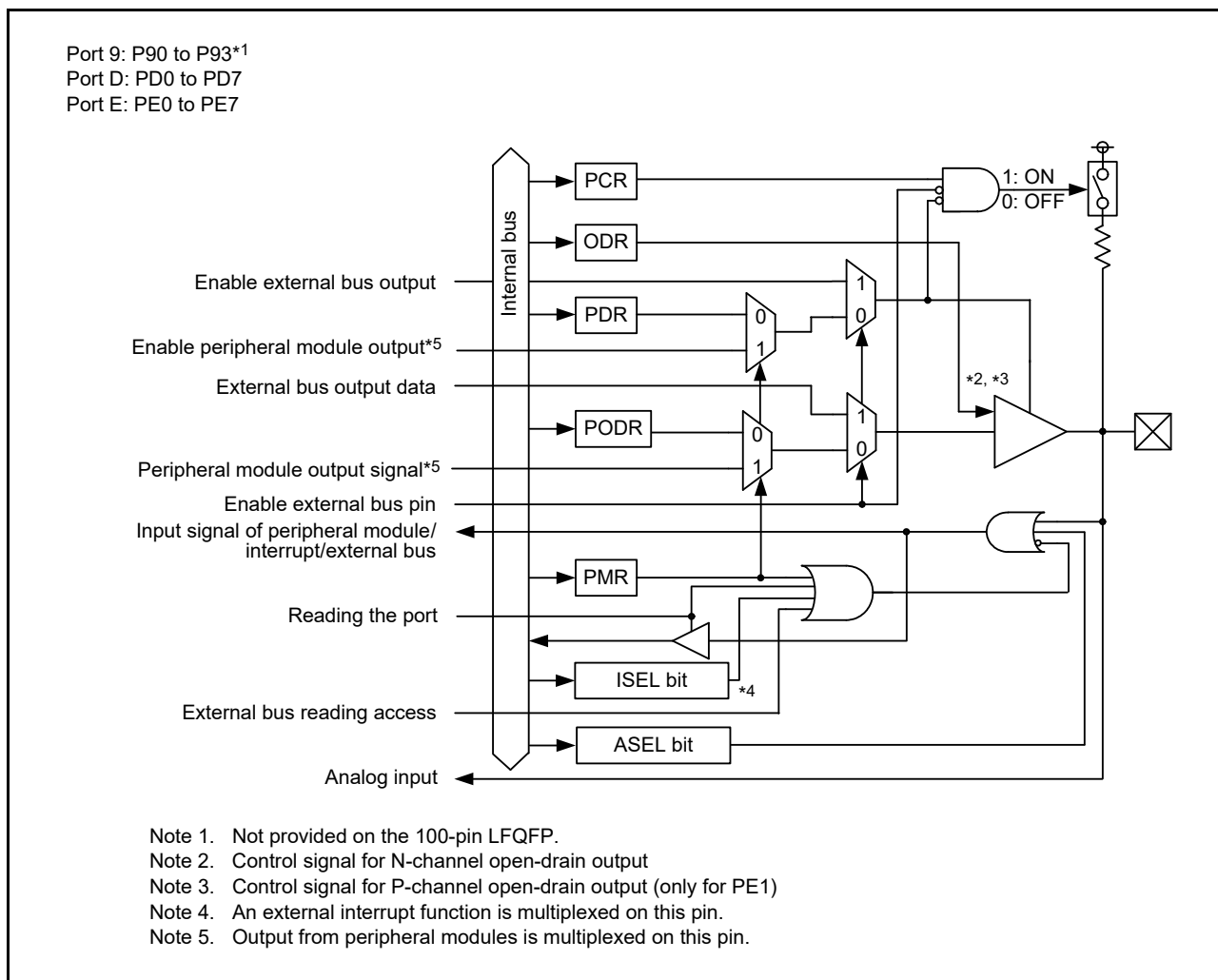


Figure 22.4 I/O Port Configuration (4)

22.3 Register Descriptions

22.3.1 Port Direction Register (PDR)

Address(es): PORT0.PDR 0008 C000h, PORT1.PDR 0008 C001h, PORT2.PDR 0008 C002h, PORT3.PDR 0008 C003h, PORT4.PDR 0008 C004h, PORT5.PDR 0008 C005h, PORT6.PDR 0008 C006h, PORT7.PDR 0008 C007h, PORT8.PDR 0008 C008h, PORT9.PDR 0008 C009h, PORTA.PDR 0008 C00Ah, PORTB.PDR 0008 C00Bh, PORTC.PDR 0008 C00Ch, PORTD.PDR 0008 C00Dh, PORTE.PDR 0008 C00Eh, PORTF.PDR 0008 C00Fh, PORTG.PDR 0008 C010h, PORTH.PDR 0008 C011h, PORTJ.PDR 0008 C012h, PORTK.PDR 0008 C013h, PORTL.PDR 0008 C014h, PORTM.PDR 0008 C015h, PORTN.PDR 0008 C016h, PORTQ.PDR 0008 C017h

b7	b6	b5	b4	b3	b2	b1	b0
B7	B6	B5	B4	B3	B2	B1	B0

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Bit Name	Description	R/W
b0	B0	Pm0 I/O Select	0: Input (Functions as an input pin.) 1: Output (Functions as an output pin.)	R/W
b1	B1	Pm1 I/O Select		R/W
b2	B2	Pm2 I/O Select		R/W
b3	B3	Pm3 I/O Select		R/W
b4	B4	Pm4 I/O Select		R/W
b5	B5	Pm5 I/O Select		R/W
b6	B6	Pm6 I/O Select		R/W
b7	B7	Pm7 I/O Select		R/W

m = 0 to 9, A to H, J to N, Q

PDR is a register which is used to select the input or output direction for individual pins of the corresponding port when the pins are configured as the general I/O pins.

Each bit of PORTm.PDR corresponds to each pin of port m; I/O direction can be specified in 1-bit units.

However, the bits that correspond to port m on the 224-pin product but do not exist on a product with fewer pins are reserved. When writing, write 1 (output) to these bits.

Each bit of PDR corresponding to port m that does not exist is reserved. Make settings according to the description in section 22.4, Initialization of the Port Direction Register (PDR).

The B5 bit in PORT3.PDR is reserved, because the P35 pin is input only.

22.3.2 Port Output Data Register (PODR)

Address(es): PORT0.PODR 0008 C020h, PORT1.PODR 0008 C021h, PORT2.PODR 0008 C022h, PORT3.PODR 0008 C023h, PORT4.PODR 0008 C024h, PORT5.PODR 0008 C025h, PORT6.PODR 0008 C026h, PORT7.PODR 0008 C027h, PORT8.PODR 0008 C028h, PORT9.PODR 0008 C029h, PORTA.PODR 0008 C02Ah, PORTB.PODR 0008 C02Bh, PORTC.PODR 0008 C02Ch, PORTD.PODR 0008 C02Dh, PORTE.PODR 0008 C02Eh, PORTF.PODR 0008 C02Fh, PORTG.PODR 0008 C030h, PORTH.PODR 0008 C031h, PORTJ.PODR 0008 C032h, PORTK.PODR 0008 C033h, PORTL.PODR 0008 C034h, PORTM.PODR 0008 C035h, PORTN.PODR 0008 C036h, PORTQ.PODR 0008 C037h

b7	b6	b5	b4	b3	b2	b1	b0
B7	B6	B5	B4	B3	B2	B1	B0

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Bit Name	Description	R/W
b0	B0	Pm0 Output Data Store	0: Low output 1: High output	R/W
b1	B1	Pm1 Output Data Store		R/W
b2	B2	Pm2 Output Data Store		R/W
b3	B3	Pm3 Output Data Store		R/W
b4	B4	Pm4 Output Data Store		R/W
b5	B5	Pm5 Output Data Store		R/W
b6	B6	Pm6 Output Data Store		R/W
b7	B7	Pm7 Output Data Store		R/W

m = 0 to 9, A to H, J to N, Q

PODR is a register which holds the data to be output from the pins used for general I/O.

Bits that correspond to port m on the 224-pin product but do not exist on a product with fewer pins are reserved. When writing, write 0 (low output) to these bits.

The B5 bit in PORT3.PODR is reserved, because the P35 pin is input only. Data is not output from the corresponding pins even if these bits are set.

The bit corresponding to a pin that does not exist is reserved. A reserved bit is read as 0. The write value should be 0.

22.3.3 Port Input Register (PIDR)

Address(es): PORT0.PIDR 0008 C040h, PORT1.PIDR 0008 C041h, PORT2.PIDR 0008 C042h, PORT3.PIDR 0008 C043h, PORT4.PIDR 0008 C044h, PORT5.PIDR 0008 C045h, PORT6.PIDR 0008 C046h, PORT7.PIDR 0008 C047h, PORT8.PIDR 0008 C048h, PORT9.PIDR 0008 C049h, PORTA.PIDR 0008 C04Ah, PORTB.PIDR 0008 C04Bh, PORTC.PIDR 0008 C04Ch, PORTD.PIDR 0008 C04Dh, PORTE.PIDR 0008 C04Eh, PORTF.PIDR 0008 C04Fh, PORTG.PIDR 0008 C050h, PORTH.PIDR 0008 C051h, PORTJ.PIDR 0008 C052h, PORTK.PIDR 0008 C053h, PORTL.PIDR 0008 C054h, PORTM.PIDR 0008 C055h, PORTN.PIDR 0008 C056h, PORTQ.PIDR 0008 C057h

b7	b6	b5	b4	b3	b2	b1	b0
B7	B6	B5	B4	B3	B2	B1	B0

Value after reset: x x x x x x x x

x: Undefined

Bit	Symbol	Bit Name	Description	R/W
b0	B0	Pm0	0: Low input 1: High input	R
b1	B1	Pm1		R
b2	B2	Pm2		R
b3	B3	Pm3		R
b4	B4	Pm4		R
b5	B5	Pm5		R
b6	B6	Pm6		R
b7	B7	Pm7		R

m = 0 to 9, A to H, J to N, Q

PIDR is a register which reflects individual pin states of the port.

The pin states of port m can be read with the PORTm.PIDR, regardless of the values of PORTm.PDR and PORTm.PMR.

The NMI pin state is reflected in the P35 bit. However, the states of pins when the PmnPFS.ASEL bit is set to 1 cannot be read.

The bit corresponding to a pin that does not exist is reserved. A reserved bit is read as undefined, and cannot be modified.

22.3.4 Port Mode Register (PMR)

Address(es): PORT0.PMR 0008 C060h, PORT1.PMR 0008 C061h, PORT2.PMR 0008 C062h, PORT3.PMR 0008 C063h, PORT4.PMR 0008 C064h, PORT5.PMR 0008 C065h, PORT6.PMR 0008 C066h, PORT7.PMR 0008 C067h, PORT8.PMR 0008 C068h, PORT9.PMR 0008 C069h, PORTA.PMR 0008 C06Ah, PORTB.PMR 0008 C06Bh, PORTC.PMR 0008 C06Ch, PORTD.PMR 0008 C06Dh, PORTE.PMR 0008 C06Eh, PORTF.PMR 0008 C06Fh, PORTG.PMR 0008 C070h, PORTH.PMR 0008 C071h, PORTJ.PMR 0008 C072h, PORTK.PMR 0008 C073h, PORTL.PMR 0008 C074h, PORTM.PMR 0008 C075h, PORTN.PMR 0008 C076h, PORTQ.PMR 0008 C077h

b7	b6	b5	b4	b3	b2	b1	b0
B7	B6	B5	B4	B3	B2	B1	B0

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Bit Name	Description	R/W
b0	B0	Pm0 Pin Mode Control	0: Uses the pin as a general I/O pin. 1: Uses the pin as an I/O port for peripheral modules.	R/W
b1	B1	Pm1 Pin Mode Control		R/W
b2	B2	Pm2 Pin Mode Control		R/W
b3	B3	Pm3 Pin Mode Control		R/W
b4	B4	Pm4 Pin Mode Control		R/W
b5	B5	Pm5 Pin Mode Control		R/W
b6	B6	Pm6 Pin Mode Control		R/W
b7	B7	Pm7 Pin Mode Control		R/W

m = 0 to 9, A to H, J to N, Q

PMR is a register which specifies the function of the pins of the port.

Each bit of PORTm.PMR corresponds to each pin of port m; pin function can be specified in 1-bit units.

However, bits that correspond to port m on the 224-pin product but do not exist on a product with fewer pins are reserved. When writing, write 0 (general I/O port) to these bits.

The bit corresponding to a pin that does not exist is reserved. A reserved bit is read as 0. The write value should be 0.

22.3.5 Open-Drain Control Register 0 (ODR0)

Address(es): PORT0.ODR0 0008 C080h, PORT1.ODR0 0008 C082h, PORT2.ODR0 0008 C084h, PORT3.ODR0 0008 C086h, PORT4.ODR0 0008 C088h, PORT5.ODR0 0008 C08Ah, PORT6.ODR0 0008 C08Ch, PORT7.ODR0 0008 C08Eh, PORT8.ODR0 0008 C090h, PORT9.ODR0 0008 C092h, PORTA.ODR0 0008 C094h, PORTB.ODR0 0008 C096h, PORTC.ODR0 0008 C098h, PORTD.ODR0 0008 C09Ah, PORTE.ODR0 0008 C09Ch, PORTF.ODR0 0008 C09Eh, PORTG.ODR0 0008 C0A0h, PORTH.ODR0 0008 C0A2h, PORTJ.ODR0 0008 C0A4h, PORTK.ODR0 0008 C0A6h, PORTL.ODR0 0008 C0A8h, PORTM.ODR0 0008 C0AAh, PORTN.ODR0 0008 C0ACh, PORTQ.ODR0 0008 C0AEh

b7	b6	b5	b4	b3	b2	b1	b0
—	B6	—	B4	B3	B2	—	B0

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Bit Name	Description	R/W
b0	B0	Pm0 Output Type Select	For pins other than the port PE1 pin	R/W
b1	—	Reserved	Odd Even bit bit	R/W
b2	B2	Pm1 Output Type Select	x 0: CMOS output	R/W
b3	B3*1	PE1 Output Type Select	x 1: N-channel open-drain output (b1, b3, b5, b7: Reserved)	R/W
b4	B4	Pm2 Output Type Select		R/W
b5	—	Reserved	For port PE1 pin b3 b2	R/W
b6	B6	Pm3 Output Type Select	0 0: CMOS output	R/W
b7	—	Reserved	0 1: N-channel open-drain output 1 0: P-channel open-drain output 1 1: Setting prohibited	R/W

m = 0 to 9, A to H, J to N, Q

Note 1. The bit for a pin other than PE1 is reserved.

ODR0 is a register which is used to select an output type for the pins of the port.

In the registers other than PORTE.ODR0, the odd bits (b1, b3, b5, and b7) are reserved.

However, the output type of the port PE1 pin is specified by the combination of b3 and b2.

Bits that correspond to port m on the 224-pin product but do not exist on a product with fewer pins are reserved. When writing, write 0 (CMOS output) to these bits.

The bit corresponding to a pin that does not exist is also reserved. A reserved bit is read as 0. The write value should be 0.

22.3.6 Open-Drain Control Register 1 (ODR1)

Address(es): PORT0.ODR1 0008 C081h, PORT1.ODR1 0008 C083h, PORT2.ODR1 0008 C085h, PORT3.ODR1 0008 C087h, PORT4.ODR1 0008 C089h, PORT5.ODR1 0008 C08Bh, PORT6.ODR1 0008 C08Dh, PORT7.ODR1 0008 C08Fh, PORT8.ODR1 0008 C091h, PORT9.ODR1 0008 C093h, PORTA.ODR1 0008 C095h, PORTB.ODR1 0008 C097h, PORTC.ODR1 0008 C099h, PORTD.ODR1 0008 C09Bh, PORTE.ODR1 0008 C09Dh, PORTF.ODR1 0008 C09Fh, PORTG.ODR1 0008 C0A1h, PORTH.ODR1 0008 C0A3h, PORTJ.ODR1 0008 C0A5h, PORTK.ODR1 0008 C0A7h, PORTL.ODR1 0008 C0A9h, PORTM.ODR1 0008 C0ABh, PORTN.ODR1 0008 C0ADh, PORTQ.ODR1 0008 C0AFh

b7	b6	b5	b4	b3	b2	b1	b0
—	B6	—	B4	—	B2	—	B0

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Bit Name	Description	R/W
b0	B0	Pm4 Output Type Select	0: CMOS output 1: N-channel open-drain output	R/W
b1	—	Reserved		R/W
b2	B2	Pm5 Output Type Select		R/W
b3	—	Reserved		R/W
b4	B4	Pm6 Output Type Select		R/W
b5	—	Reserved		R/W
b6	B6	Pm7 Output Type Select		R/W
b7	—	Reserved		R/W

m = 0 to 9, A to H, J to N, Q

ODR1 is used to select an output type for each pin of the port.

The odd bits (b1, b3, b5, and b7) in the ODR1 register are reserved.

Bits that correspond to port m on the 224-pin product but do not exist on a product with fewer pins are reserved. When writing, write 0 (CMOS output) to these bits.

The bit corresponding to a pin that does not exist is also reserved. A reserved bit is read as 0. The write value should be 0.

22.3.7 Pull-Up Resistor Control Register (PCR)

Address(es): PORT0.PCR 0008 C0C0h, PORT1.PCR 0008 C0C1h, PORT2.PCR 0008 C0C2h, PORT3.PCR 0008 C0C3h, PORT4.PCR 0008 C0C4h, PORT5.PCR 0008 C0C5h, PORT6.PCR 0008 C0C6h, PORT7.PCR 0008 C0C7h, PORT8.PCR 0008 C0C8h, PORT9.PCR 0008 C0C9h, PORTA.PCR 0008 C0CAh, PORTB.PCR 0008 C0CBh, PORTC.PCR 0008 C0CCh, PORTD.PCR 0008 C0CDh, PORTE.PCR 0008 C0CEh, PORTF.PCR 0008 C0CFh, PORTG.PCR 0008 C0D0h, PORTH.PCR 0008 C0D1h, PORTJ.PCR 0008 C0D2h, PORTK.PCR 0008 C0D3h, PORTL.PCR 0008 C0D4h, PORTM.PCR 0008 C0D5h, PORTN.PCR 0008 C0D6h, PORTQ.PCR 0008 C0D7h

b7	b6	b5	b4	b3	b2	b1	b0
B7	B6	B5	B4	B3	B2	B1	B0

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Bit Name	Description	R/W
b0	B0	Pm0 Input Pull-Up Resistor Control	0: Disables an input pull-up resistor. 1: Enables an input pull-up resistor.	R/W
b1	B1	Pm1 Input Pull-Up Resistor Control		R/W
b2	B2	Pm2 Input Pull-Up Resistor Control		R/W
b3	B3	Pm3 Input Pull-Up Resistor Control		R/W
b4	B4	Pm4 Input Pull-Up Resistor Control		R/W
b5	B5	Pm5 Input Pull-Up Resistor Control		R/W
b6	B6	Pm6 Input Pull-Up Resistor Control		R/W
b7	B7	Pm7 Input Pull-Up Resistor Control		R/W

m = 0 to 9, A to H, J to N, Q

PCR is a register which enables or disables an input pull-up resistor for individual pins of the port.

While a pin is in the input state with the corresponding bit in PORTm.PCR set to 1, the pull-up resistor connected to the pin is enabled.

When a pin is set as an external bus pin (except for the WAIT pin), a general port output pin, or a peripheral module output pin, the pull-up resistor for the pin is disabled regardless of the settings of PCR.

However, when a pin is used as an address bus or bus control signal and the MCU transitions to software standby mode or deep software standby mode while the output port enable bit (SBYCR.OPE) in the standby control register is 0, the value in the PCR register becomes enabled.

The pull-up resistor is also disabled in the reset state.

The other bits are also reserved because they correspond to pins that do not exist is reserved. A reserved bit is read as 0.

The write value should be 0.

22.3.8 Drive Capacity Control Register (DSCR)

Address(es): PORT0.DSCR 0008 C0E0h, PORT1.DSCR 0008 C0E1h, PORT2.DSCR 0008 C0E2h, PORT5.DSCR 0008 C0E5h, PORT7.DSCR 0008 C0E7h, PORT8.DSCR 0008 C0E8h, PORT9.DSCR 0008 C0E9h, PORTA.DSCR 0008 C0EAh, PORTB.DSCR 0008 C0EBh, PORTC.DSCR 0008 C0ECh, PORTD.DSCR 0008 C0EDh, PORTE.DSCR 0008 C0EEh, PORTG.DSCR 0008 C0F0h, PORTH.DSCR 0008 C0F1h, PORTJ.DSCR 0008 C0F2h, PORTK.DSCR 0008 C0F3h, PORTL.DSCR 0008 C0F4h, PORTM.DSCR 0008 C0F5h, PORTN.DSCR 0008 C0F6h, PORTQ.DSCR 0008 C0F7h

b7	b6	b5	b4	b3	b2	b1	b0
B7	B6	B5	B4	B3	B2	B1	B0

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Bit Name	Description	R/W
b0	B0	Pm0 Drive Capacity Control	0: Normal drive output 1: High-drive output	R/W
b1	B1	Pm1 Drive Capacity Control		R/W
b2	B2	Pm2 Drive Capacity Control		R/W
b3	B3	Pm3 Drive Capacity Control		R/W
b4	B4	Pm4 Drive Capacity Control		R/W
b5	B5	Pm5 Drive Capacity Control		R/W
b6	B6	Pm6 Drive Capacity Control		R/W
b7	B7	Pm7 Drive Capacity Control		R/W

m = 0 to 2, 5, 7 to 9, A to E, G, H, J to N, Q

DSCR is a register which is used to switch the drive capacity of the port.

When pins are set for high-speed interface high-drive in the DSCR2 register, the drive capacity cannot be changed. For setting of the drive capacity by the DSCR and DSCR2 registers, refer to Table 22.3, Drive Capacity Setting by DSCR and DSCR2 Registers.

The bit corresponding to a pin whose drive capacity cannot be switched is readable and writable, but the drive capacity cannot be changed.

The bit corresponding to a pin that does not exist is reserved. A reserved bit is read as 0. The write value should be 0.

22.3.9 Drive Capacity Control Register 2 (DSCR2)

Address(es): PORT0.DSCR2 0008 C128h, PORT1.DSCR2 0008 C129h, PORT2.DSCR2 0008 C12Ah, PORT3.DSCR2 0008 C12Bh, PORT5.DSCR2 0008 C12Dh, PORT6.DSCR2 0008 C12Eh, PORT7.DSCR2 0008 C12Fh, PORT8.DSCR2 0008 C130h, PORT9.DSCR2 0008 C131h, PORTA.DSCR2 0008 C132h, PORTB.DSCR2 0008 C133h, PORTC.DSCR2 0008 C134h, PORTD.DSCR2 0008 C135h, PORTE.DSCR2 0008 C136h, PORTG.DSCR2 0008 C138h, PORTH.DSCR2 0008 C139h, PORTJ.DSCR2 0008 C13Ah, PORTK.DSCR2 0008 C13Bh, PORTL.DSCR2 0008 C13Ch, PORTM.DSCR2 0008 C13Dh, PORTN.DSCR2 0008 C13Eh, PORTQ.DSCR2 0008 C13Fh

b7	b6	b5	b4	b3	b2	b1	b0
B7	B6	B5	B4	B3	B2	B1	B0

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Bit Name	Description	R/W
b0	B0	Pm0 Drive Capacity Control 2	0: Normal drive/high-drive output*1 1: High-speed interface high-drive output	R/W
b1	B1	Pm1 Drive Capacity Control 2		R/W
b2	B2	Pm2 Drive Capacity Control 2		R/W
b3	B3	Pm3 Drive Capacity Control 2		R/W
b4	B4	Pm4 Drive Capacity Control 2		R/W
b5	B5	Pm5 Drive Capacity Control 2		R/W
b6	B6	Pm6 Drive Capacity Control 2		R/W
b7	B7	Pm7 Drive Capacity Control 2		R/W

m = 0 to 3, 5, 7 to 9, A to E, G, H, J to N, Q

Note 1. Pins that support switching drive capacity by the DSCR register depend on the setting of the DSCR register.

DSCR2 is a register which is used to switch the drive capacity of the port.

Table 22.3 lists the setting of drive capacity by the DSCR and DSCR2 registers.

The bit corresponding to a pin whose drive capacity cannot be switched is readable and writable, but the drive capacity cannot be changed.

The bit corresponding to a pin that does not exist is reserved. A reserved bit is read as 0. The write value should be 0.

Set the register only based on an instruction in the application guide.

Table 22.3 Drive Capacity Setting by DSCR and DSCR2 Registers

PORTm.DSCR2.Bx	PORTm.DSCR.Bx	Drive Capacity*1
0	0	Normal drive output
0	1	High-drive output
1	Don't care	High-speed interface high-drive output

Note 1. When drive capacity is fixed, or drive capacity of a pin whose drive capacity cannot be switched cannot be changed.

22.4 Initialization of the Port Direction Register (PDR)

Initialize reserved bits in the PDR register according to Table 22.4 to Table 22.7.

- The blank columns in Table 22.4 to Table 22.7 indicate the bits corresponding to the pins listed in Table 22.1, Specifications of I/O Ports.

The corresponding bits should be set to 1 (output) or 0 (input) depending on the user system.

However, the PORT3.PDR.B5 bit of the input-only P35 pin is reserved.

This bit should be set to 0 (input).

- The columns other than the blank columns in Table 22.4 to Table 22.7 indicate reserved bits. A reserved bit should be set to 0 (input) or 1 (output) according to Table 22.4 to Table 22.7. When setting a value to a reserved bit, access in byte units.

Table 22.4 PDR Register Settings in 224-Pin Packages

Port Symbol	PDR Register							
	b7	b6	b5	b4	b3	b2	b1	b0
PORT0		1		1				
PORT1								
PORT2								
PORT3			0					
PORT4								
PORT5								
PORT6								
PORT7								
PORT8								
PORT9								
PORTA								
PORTB								
PORTC								
PORTD								
PORTE								
PORTF	1	1						
PORTG								
PORTH								
PORTJ	1	1		1				
PORTK								
PORTL								
PORTM								
PORTN	1	1						
PORTQ								

Table 22.5 PDR Register Settings in 176-Pin Packages

Port Symbol	PDR Register							
	b7	b6	b5	b4	b3	b2	b1	b0
PORT0		1		1				
PORT1								
PORT2								
PORT3			0					
PORT4								
PORT5								
PORT6								
PORT7								
PORT8								
PORT9								
PORTA								
PORTB								
PORTC								
PORTD								
PORTE								
PORTF	1	1						
PORTG								
PORTH	1	1	1	1	1	1	1	1
PORTJ	1	1		1				
PORTK	1	1	1	1	1	1	1	1
PORTL	1	1	1	1	1	1	1	1
PORTM	1	1	1	1	1	1	1	1
PORTN	1	1	1	1	1	1	1	1
PORTQ	1	1	1	1	1	1	1	1

Table 22.6 PDR Register Settings in 145-pin and 144-Pin Packages

Port Symbol	PDR Register							
	b7	b6	b5	b4	b3	b2	b1	b0
PORT0		1		1				
PORT1							1	1
PORT2								
PORT3			0					
PORT4								
PORT5	1							
PORT6								
PORT7								
PORT8			1	1				
PORT9	1	1	1	1				
PORTA								
PORTB								
PORTC								
PORTD								
PORTE								
PORTF	1	1		1	1	1	1	1
PORTG	1	1	1	1	1	1	1	1
PORTH	1	1	1	1	1	1	1	1
PORTJ	1	1		1		1	1	1
PORTK	1	1	1	1	1	1	1	1
PORTL	1	1	1	1	1	1	1	1
PORTM	1	1	1	1	1	1	1	1
PORTN	1	1	1	1	1	1	1	1
PORTQ	1	1	1	1	1	1	1	1

Table 22.7 PDR Register Settings in 100-Pin Packages

Port Symbol	PDR Register							
	b7	b6	b5	b4	b3	b2	b1	b0
PORT0		1		1	1	1	1	1
PORT1							1	1
PORT2								
PORT3			0					
PORT4								
PORT5	1	1						
PORT6	1	1	1	1	1	1	1	1
PORT7	1	1	1	1	1	1	1	1
PORT8	1	1	1	1	1	1	1	1
PORT9	1	1	1	1	1	1	1	1
PORTA								
PORTB								
PORTC								
PORTD								
PORTE								
PORTF	1	1	1	1	1	1	1	1
PORTG	1	1	1	1	1	1	1	1
PORTH	1	1	1	1	1	1	1	1
PORTJ	1	1	1	1		1	1	1
PORTK	1	1	1	1	1	1	1	1
PORTL	1	1	1	1	1	1	1	1
PORTM	1	1	1	1	1	1	1	1
PORTN	1	1	1	1	1	1	1	1
PORTQ	1	1	1	1	1	1	1	1

22.5 Handling of Unused Pins

Details on the handling of unused pins are given in Table 22.8.

Table 22.8 Handling of Unused Pins

Pin Name	Handling
EMLE	Connect this pin to VSS via a resistor (pulling down).
BSCANP	Connect this pin to VSS via a resistor (pulling down).
MD	Use this as a mode pin.
RES#	Connect this pin to VCC via a resistor (pulling up).
VCC_USB	Connect this pin to VCC
VSS_USB	Connect this pin to VSS
USB0_DP	Keep these pins open.
USB0_DM	
P35/NMI	Connect this pin to VCC via a resistor (pulling up).
P36/EXTAL	Set the MOSCCR.MOSTP bit to 1 (the main clock oscillator is stopped) when not using the main clock When this pin is not used as port P36, handle as port 0 to port 9, port A to port H, port J to port N, port Q.
P37/XTAL	Set the MOSCCR.MOSTP bit to 1 (the main clock oscillator is stopped) when not using the main clock When this pin is not used as port P37, handle as port 0 to port 9, port A to port H, port J to port N, port Q When an external clock is input to the EXTAL pin, keep the pin open.
XCIN	Connect this pin to VSS via a resistor (pulling down).
XCOUT	Keep this pin open.
Port 0 to Port 9, Port A to Port H, Port J to Port N, Port Q	- If the direction setting is for input (PORTn.PDR = 0), pins are connected to VCC (pulled up) via resistors or to VSS (pulled down) via resistors, pin by pin.*¹ - If the direction setting is for output (PORTn.PDR = 1), the pin is released.*¹, *²
VREFH0	Connect this pin to AVCC0.
VREFL0	Connect this pin to AVSS0.
AVCC0	Connect this pin to VCC when not using the 12-bit A/D converter (unit 0).
AVSS0	Connect this pin to VSS when not using the 12-bit A/D converter (unit 0).
AVCC1	Connect this pin to VCC when not using the 12-bit A/D converter (unit 1) and D/A converter.
AVSS1	Connect this pin to VSS when not using the 12-bit A/D converter (unit 1) and D/A converter.

Note 1. Set the PORTn.PMR bit, the PmnPFS.ISEL bit and the PmnPFS.ASEL bit to 0.

Note 2. In the case of release when the setting is for output, the port is an input over the period from release from the reset state to the pin becoming an output. Since the voltage on the pin is undefined while it is an input, this may lead to an increase in the current drawn.